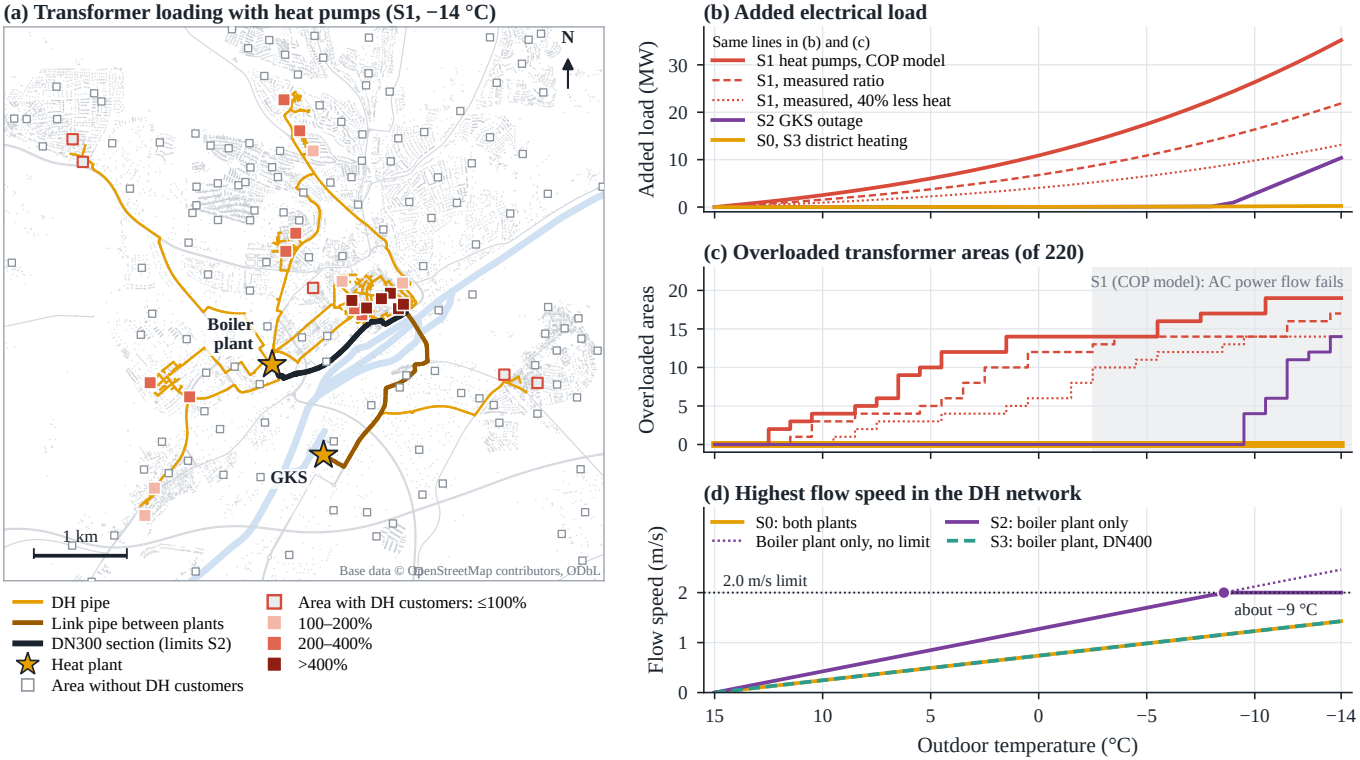

(a) Transformer loading with heat pumps (S1, −14 °C)
N
Boiler plant
GKS
1 km
Base data © OpenStreetMap contributors, ODbL
DH pipe
Link pipe between plants
DN300 section (limits S2)
Heat plant
Area without DH customers
Area with DH customers: ≤100%
100–200%
200–400%
>400%
(b) Added electrical load
Same lines in (b) and (c)
S1 heat pumps, COP model
S1, measured ratio
S1, measured, 40% less heat
S2 GKS outage
S0, S3 district heating
Added load (MW)
(c) Overloaded transformer areas (of 220)
S1 (COP model): AC power flow fails
Overloaded areas
(d) Highest flow speed in the DH network
S0: both plants
Boiler plant only, no limit
S2: boiler plant only
S3: boiler plant, DN400
2.0 m/s limit
about −9 °C
Flow speed (m/s)
Outdoor temperature (°C)